25901177 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCMA200 - DATA ANALYTICS USING R

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define the repeat loop in R. [U] / [CO1]
2. Explain the purpose of the dir() function in R. [K] / [CO2]
3. Write any two advantages of a 3D pie chart over the bar chart. [U] / [CO3]
4. Recall the file extension of an Excel file? [K] / [CO2]
5. Write the output of the code: data <- c(10, 20, 30, 40, 50) mean(data). [A] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Compare the if–else if ladder and switch statement (any five points). [U] / [CO1]
7. Explain the concept of melting a data frame. [K] / [CO2]

8. Compare a scatter plot, a whisker plot, and a box plot. [U] / [CO3]

9. Explain vectors in R with suitable examples. [U] / [CO1]

10. Create a data frame product with ProductID, ProductName, and Price. Write R commands to: a) display the first five rows, b) display the last five rows. [U] / [CO2]

11. Compare median and mode in real-life decision making with suitable examples. [An] / [CO3]

12. What is the use of the str() function? Explain with an example. [U] / [CO2]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Explain the concept of row binding in R using rbind().Also Create two data frames with identical column names representing sales data of two different months. Combine them using rbind() [U] / [CO2]

14. With suitable R code, explain how to create and customize a pie chart using base R. [A] / [CO3]

[15x1 = 15]